



LOYOLA INTERNATIONAL SCHOOL

Qatar

Academic Session 2026 – 27

Worksheet

Grade:6

Subject: Mathematics

Date:

Lesson / Topic: Chapter:2-Lines and Angles

1. Geometric Foundations

1. Match each real-world example to the correct geometric term from the word bank. Write the term in the table.

Word bank: Point, Line, Line Segment, Ray

Image	Real-World Example	Geometric Term
	A beam of light from a laser pointer	
	A straight highway that extends forever in both directions	
	A new, unsharpened pencil	
	A single star in the night sky	

2. Angles in Motion 🤖



2. Making Rotating Arms: In a science project, Ahmed builds a mechanical robot arm that rotates. The arm starts flat against the table (0°). It rotates exactly $\frac{1}{4}$ of a full circle upwards. What is the measure of the angle formed? What special type of angle is this?

3. Comparing Angles: Fatima has a pair of scissors open slightly to cut a piece of thread. Mariam has a pair of scissors open wide to cut a large sheet of cardboard. Who has created the larger angle with their scissors? Explain your reasoning.

3. Types of Angles and Their Measures 🕒

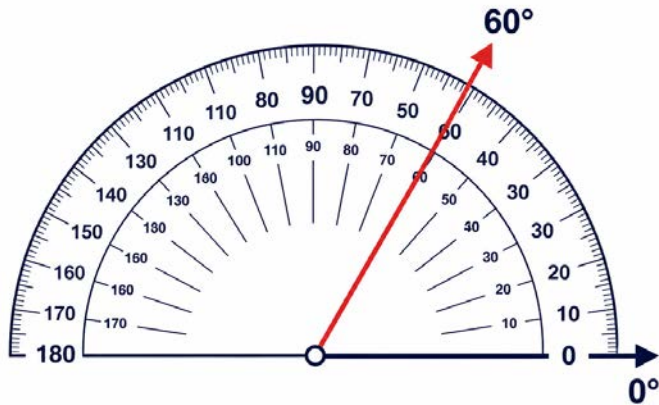
4. Look at the angle measures below. Classify each one as **Acute**, **Right**, **Obtuse**, **Straight**, or **Reflex**.

Angle Measure	Angle Type	Example Time on a Clock
90°		3:00
45°		1:30
150°		5:00

180°

6:00

4. Measuring and Drawing Angles



5. Measuring Angles: Look at the protractor diagram. What is the exact measure of the angle shown?

6. Drawing Angles: Using a ruler and a protractor, draw an acute angle that measures exactly 55° . Label the vertex 'A'.

7. Drawing Angles: Using a ruler and a protractor, draw an obtuse angle that measures exactly 135° . Label the vertex 'B'.

5. Competency Challenges 🏆

8. Ali is designing a new skate ramp in Doha. The ramp needs to meet the ground at an angle less than 90° so skaters can smoothly roll onto it. What special type of angle must this be? Give one possible measurement for the ramp's angle.

9. A laptop is open so the screen forms an angle of 110° with the keyboard. If you push the screen forward (closing it) by 30° , what is the new angle measure? What type of angle is it now?

10. In geometry class, a student draws a **Line Segment** on the board. Another student draws a **Line**. In your own words, explain the primary difference between a Line Segment and a Line.
